

Exciting assignment! Let's break down the requirements and provide a step-by-step guide to help you complete this challenging project.

**\*\*Backend Requirements (Node.js, Express)\*\***

1. Develop a REST API using Node.js and Express:

- \* Use any database of your choice (e.g., MongoDB, PostgreSQL, MySQL).
- \* Implement CRUD operations for charging stations:
  - + Create a new charging station.
  - + Read (List) all charging stations.
  - + Update a charging station.
  - + Delete a charging station.

2. Implement User Authentication using JWT:

- \* Register (Signup) API.
- \* Login API (Issue JWT token).
- \* Protected Routes for managing charging stations.

**\*\*Frontend Requirements (Vue.js)\*\***

1. Create a Login Screen that interacts with the backend API.
2. Build a Charger Listing Page:

- \* Display all onboarded chargers.
- \* Add filters (e.g., status, power output, connector type).
- \* Add/Edit/Delete chargers.

3. Develop a Map View:

- \* Display all onboarded chargers on a map using Google Maps or OpenStreetMap.

- \* Clicking a marker should show the charger details.

**\*\*Deployment Requirements\*\***

1. Deploy both the backend and frontend on any cloud platform (e.g., AWS, Heroku, Vercel, Render, or Firebase Hosting).
2. Ensure the backend has a public API endpoint accessible for testing.

**\*\*Deliverables\*\***

1. A GitHub Repository containing:
  - \* Backend code